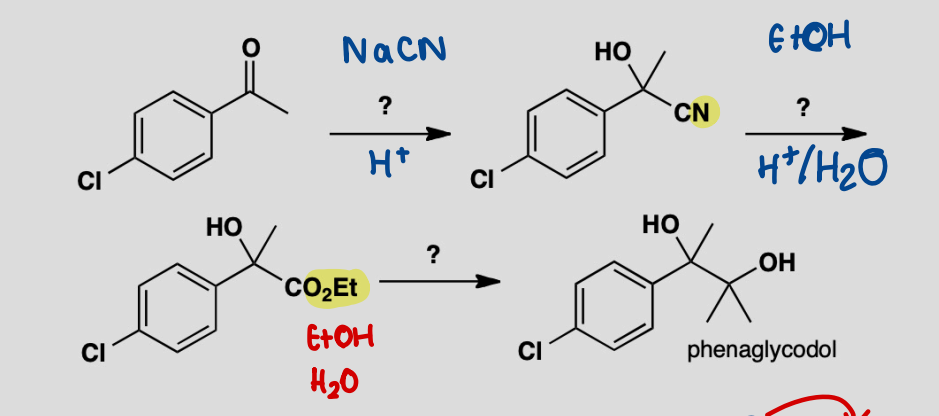
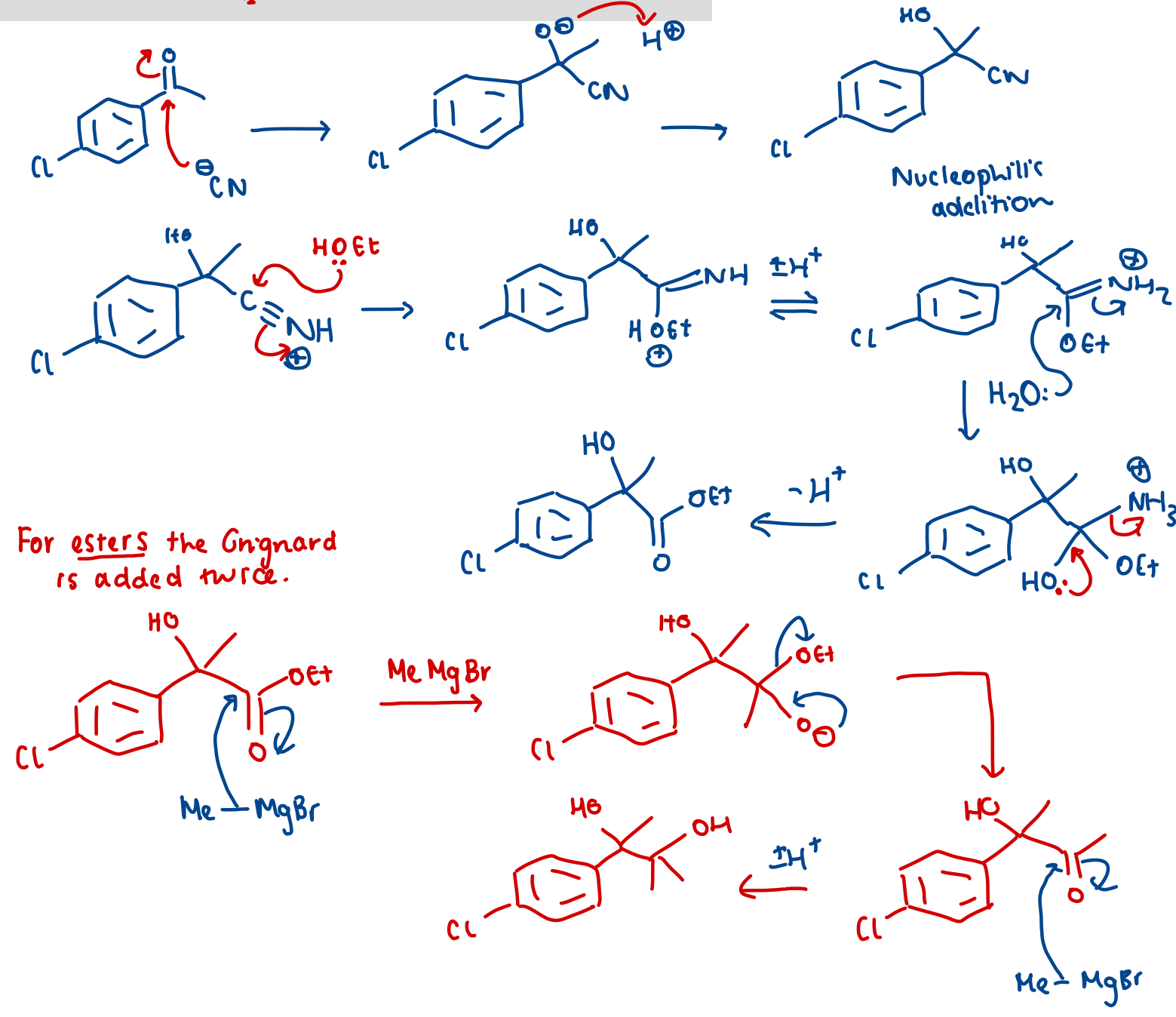


PROBLEM 1

Suggest reagents to make the drug phenaglycodol by the route below.



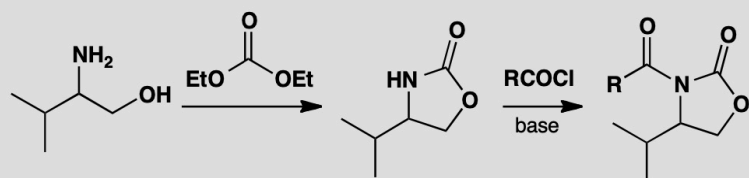
- Nucleophilic addition of CN^- to aldehyde/ketone to extend the carbon chain
- Convert nitriles to esters or carboxylic acids
- Grignard reactions
ketone/aldehyde $\rightarrow$ add once
Ester $\rightarrow$ add twice



For esters the Grignard is added twice.

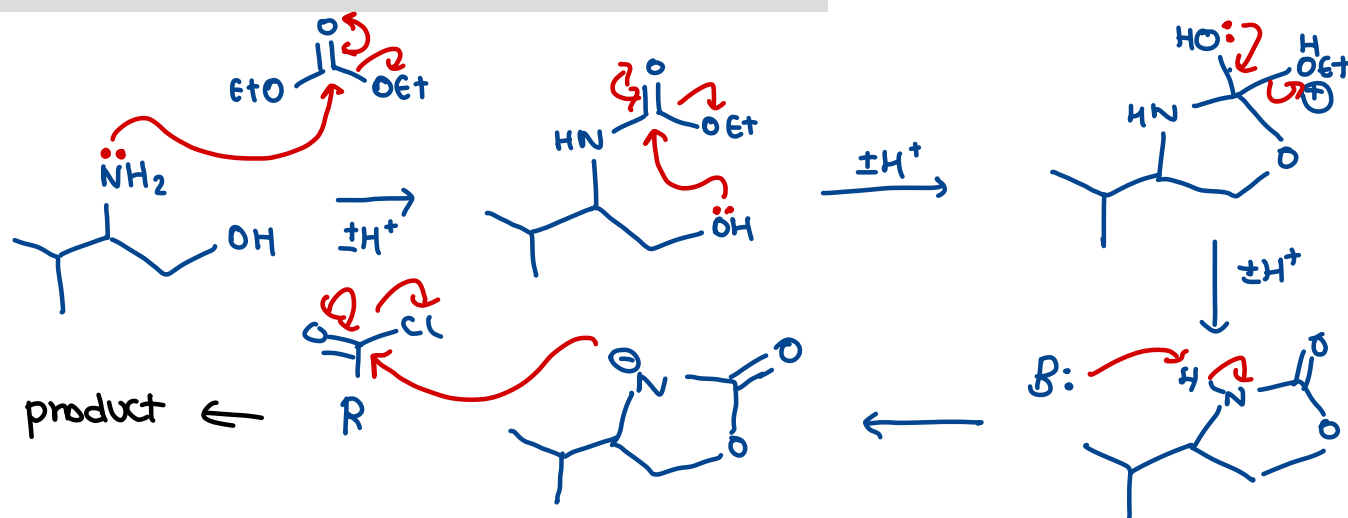
PROBLEM 4

Suggest mechanisms for these reactions.

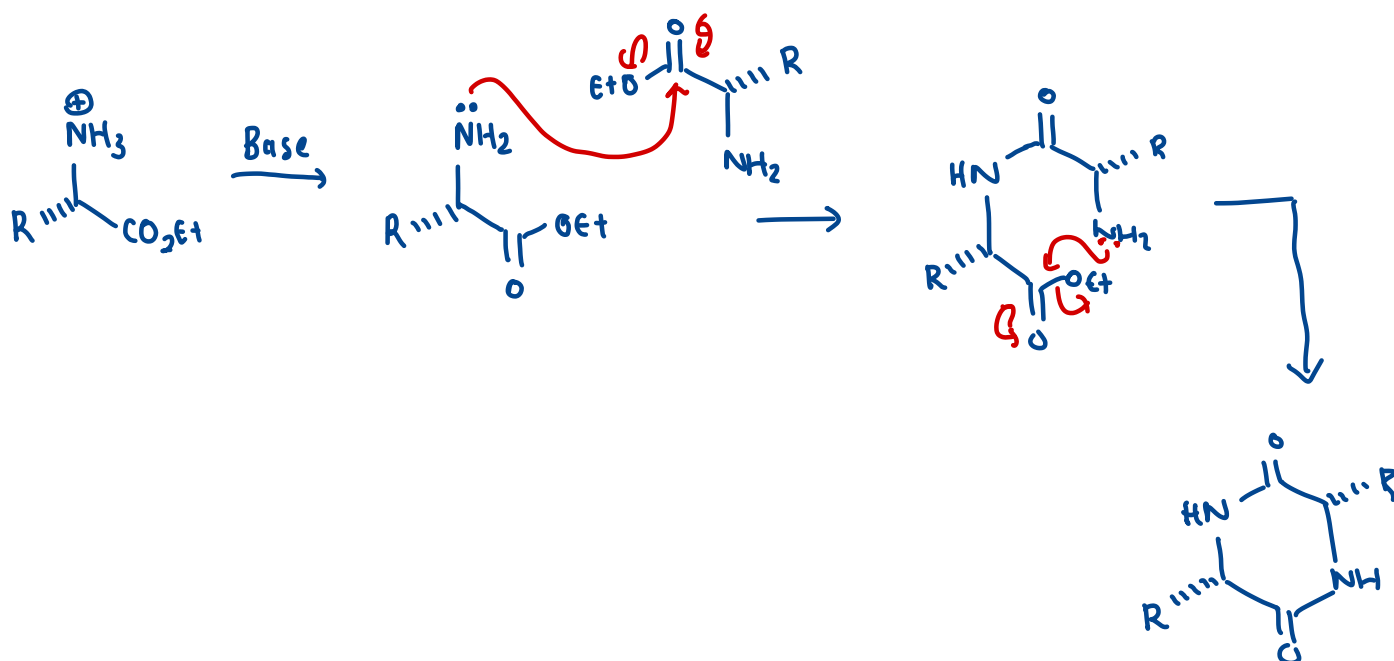
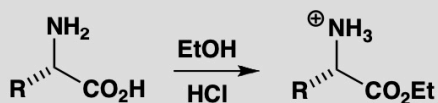


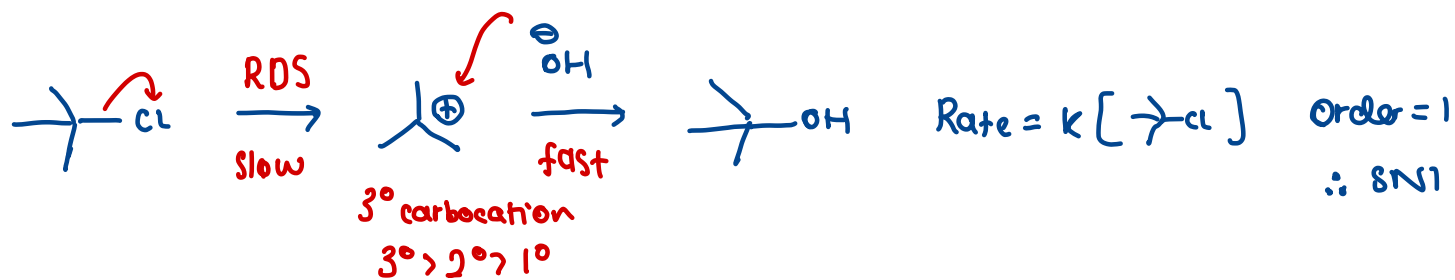
N less electronegative than O

∴ N better nucleophile because lone pair is higher energy

**PROBLEM 5**

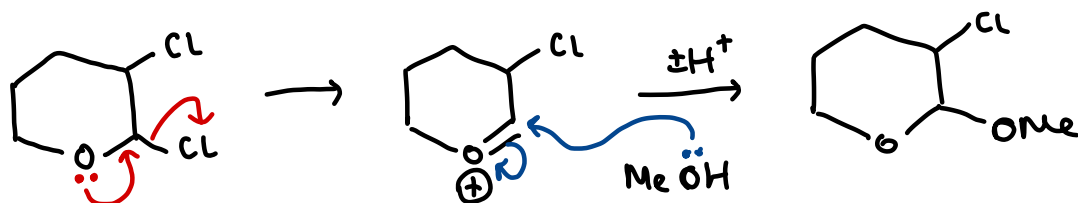
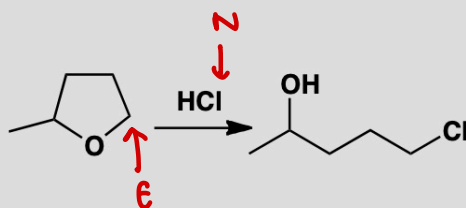
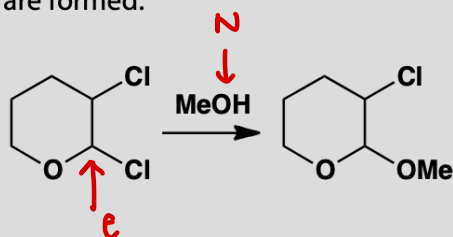
In making esters of the naturally occurring amino acids (general structure below) it is important to keep them as their hydrochloride salts. What would happen to these compounds if they were neutralized?



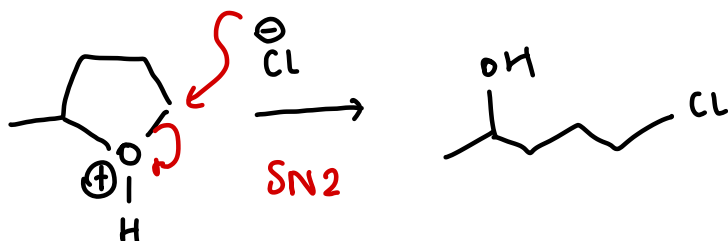


PROBLEM 3

Draw mechanisms for these reactions, explaining why these particular products are formed.

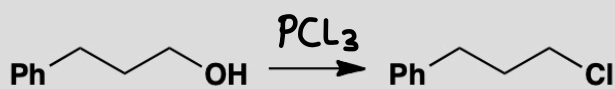
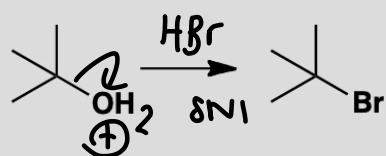


SN1



PROBLEM 4

Suggest how to carry out the following transformations.



Can't do $\text{S}_{\text{N}}2$
 OH^- not a LG

